

AccentAdapt-Whisper: Improving English ASR Performance for Asian Accents using an Adaptive Ensemble Method

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ABSTRACT

Automatic Speech Recognition systems continue to struggle with non-native English speech, especially from speakers with Asian accents. While large-scale models like OpenAI's Whisper have shown impressive generalization capabilities, they still experience significant performance drops when faced with accent variability, and region-specific fine-tuning remains computationally expensive.

We present **AccentAdapt-Whisper**, a lightweight ensemble framework that addresses this challenge through intelligent routing rather than retraining. Our method uses a CNN-based accent detector to automatically direct non-native speech to Whisper-Medium and native English to Whisper-Small, matching model capacity to task difficulty **without requiring any fine-tuning**.

17%

Relative WER Improvement (14.98% → 12.43%)

ASR Accent Adaptation Ensemble Learning Multimodal Routing Whisper L2-ARCTIC

CommonVoice

PROBLEM & MOTIVATION

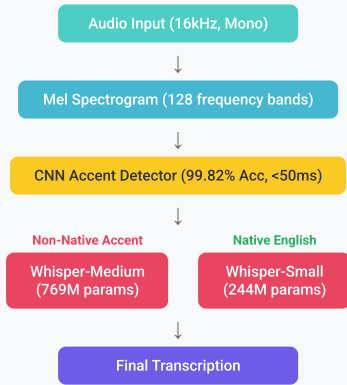
Challenge: ASR systems perform significantly worse for non-native English speakers, particularly those with Asian accents (Vietnamese, Chinese, Korean, Indian). Studies consistently show that ASR models trained primarily on native English speech struggle with phonological and prosodic variations.

Current Limitations:

- Fine-tuning requires large volumes of accent-labeled training data
- Computationally expensive model retraining and significant resources
- Post-processing pipelines correct errors reactively rather than preventing them
- No practical real-time solutions available for diverse accents

Our Solution: Intelligent routing at inference time using pre-trained models in ensemble configuration, achieving performance gains without additional training or external APIs.

SYSTEM ARCHITECTURE



Key Design Principles:

- Preprocessing:** Standardize audio to 16kHz mono, extract Mel spectrograms preserving prosodic features
- CNN Detector:** 5-class classification (Vietnamese, Chinese, Korean, Indian, Native English)
- Adaptive Routing:** Deterministic rule-based selection, no trainable parameters required
- Computational Efficiency:** Whisper-Small provides 6x faster inference for native speech

EXPERIMENTAL RESULTS

Dataset: L2-ARCTIC corpus + CommonVoice native English (2,885 utterances across 5 accent categories)

Overall WER Comparison:



Per-Accent WER Performance:

Accent	Whisper-Small	Whisper-Medium	AccentAdapt
Vietnamese	18.2%	14.8%	14.8%
Chinese	16.5%	13.2%	13.2%
Korean	15.8%	12.9%	12.9%
Indian	14.3%	11.5%	11.5%
Native English	8.1%	8.0%	8.1%

STATISTICAL VALIDATION



95% Confidence Interval for WER improvement: **2.10 - 2.98 percentage points**. Improvements statistically significant at $p < 0.05$ for each non-native accent category. The ensemble demonstrates enhanced robustness with reduced variance and fewer outliers compared to baselines.

Accent Detection Accuracy (99.82% Overall):

Category	Accuracy (%)
XN Vietnamese	99.8%
CN Chinese	99.9%
KR Korean	99.8%
IN Indian	99.7%
US Native	99.9%

KEY CONTRIBUTIONS & CONCLUSIONS

1. Novel Routing Strategy: Accent-adaptive ensemble requiring **no fine-tuning**, no external APIs, and no manual accent-specific rules

2. Significant WER Reduction: 17% relative improvement (14.98% → 12.43%) with strong statistical significance ($t=11.32$, $p<0.001$)

3. High-Accuracy Detection: 99.82% classification accuracy with minimal latency overhead (<50ms)

Feature	Traditional Approach	AccentAdapt-Whisper
No Fine-tuning Required	✗	✓
No Large Training Data Needed	✗	✓
Real-time Processing Capable	✗	✓
Modular & Extensible Design	✗	✓

Impact: This approach demonstrates that inference-level adaptation represents a practical, scalable, and fair solution for addressing accent-related performance disparities, lowering barriers to deploying more inclusive speech technology.